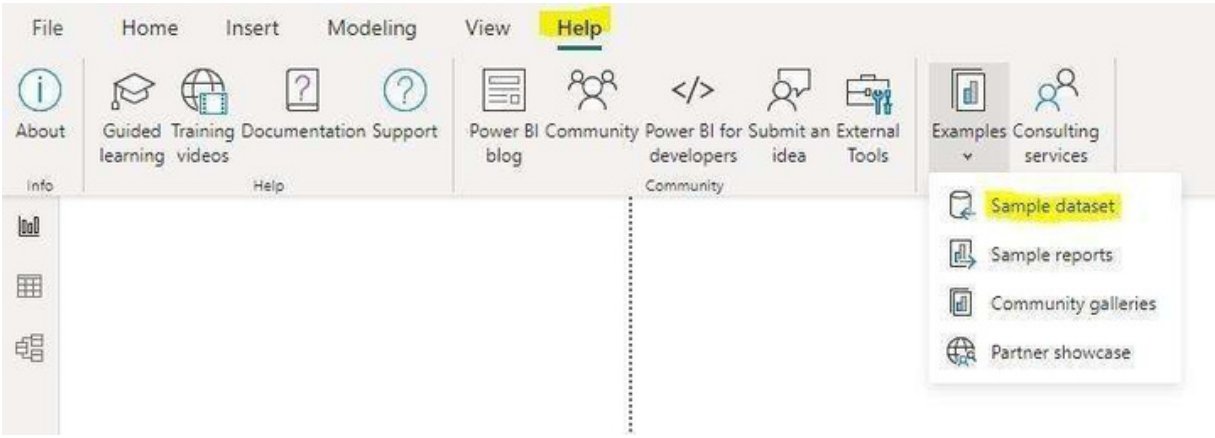
EX.NO: 5	APPLY THE WHAT – IF ANALYSIS FOR DATA VISUALIZATION TO DESIGN AND GENERATE NECESSARY REPORTS BASED ON THE DATA WAREHOUSE DATA
DATE:	

AIM:

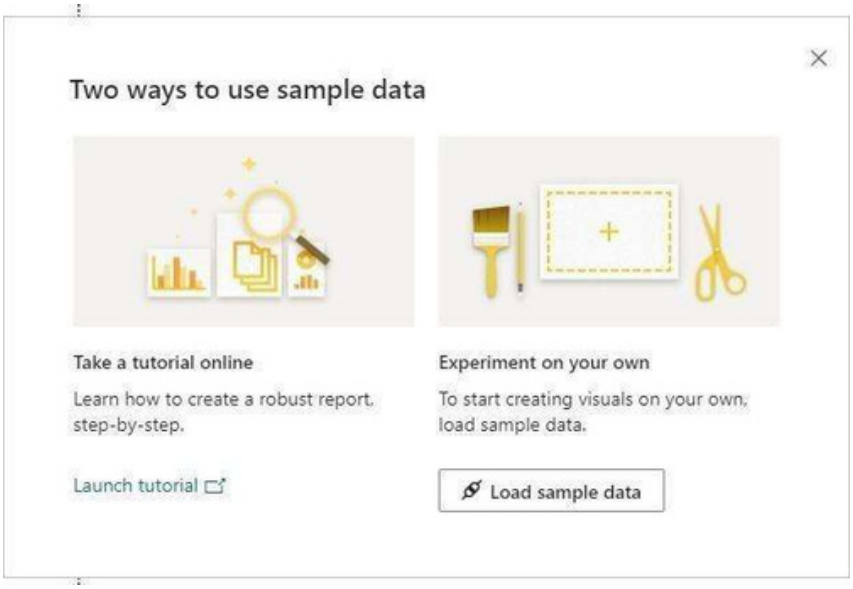
To apply the what – If analysis for Data Visualization to design and generate necessary reports based on the data warehouse data in Excel.

PROCEDURE:

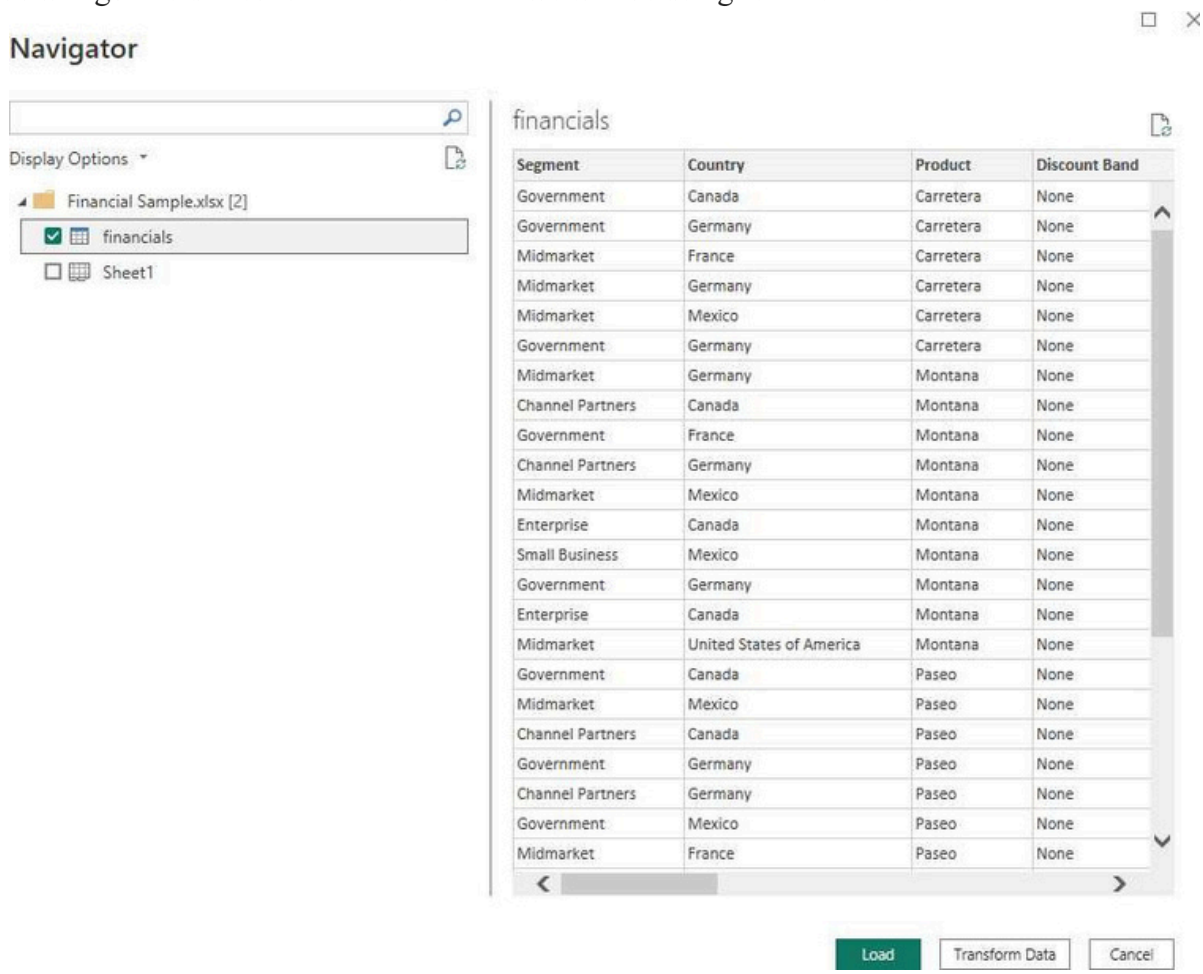
Step 1: Go to Power BI Desktop, click 'Help' , then choose the icon 'Examples' and press on 'Sample dataset'.



Step 2: The box below appears after selecting "Sample dataset." Next, select "Load sample data."



We will get the dataset of ‘financials’ that we will be using for this article.



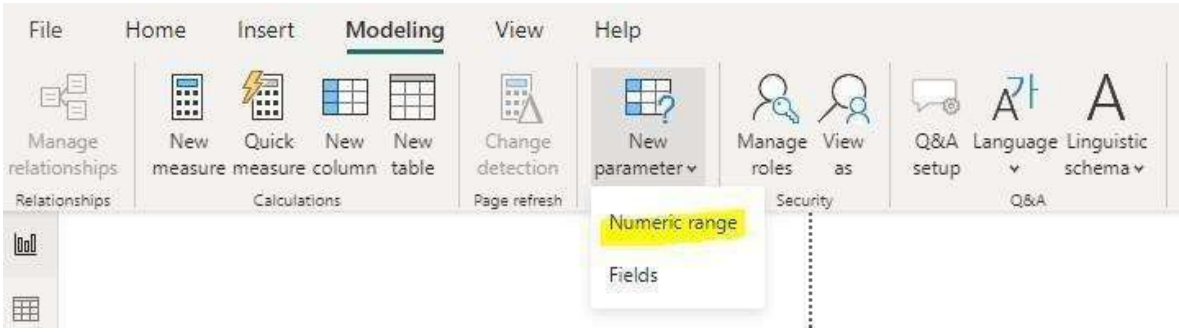
Segment	Country	Product	Discount Band
Government	Canada	Carretera	None
Government	Germany	Carretera	None
Midmarket	France	Carretera	None
Midmarket	Germany	Carretera	None
Midmarket	Mexico	Carretera	None
Government	Germany	Carretera	None
Midmarket	Germany	Montana	None
Channel Partners	Canada	Montana	None
Government	France	Montana	None
Channel Partners	Germany	Montana	None
Midmarket	Mexico	Montana	None
Enterprise	Canada	Montana	None
Small Business	Mexico	Montana	None
Government	Germany	Montana	None
Enterprise	Canada	Montana	None
Midmarket	United States of America	Montana	None
Government	Canada	Paseo	None
Midmarket	Mexico	Paseo	None
Channel Partners	Canada	Paseo	None
Government	Germany	Paseo	None
Channel Partners	Germany	Paseo	None
Government	Mexico	Paseo	None
Midmarket	France	Paseo	None

Steps for creating what-if parameters.

Step 1: Create a parameter.

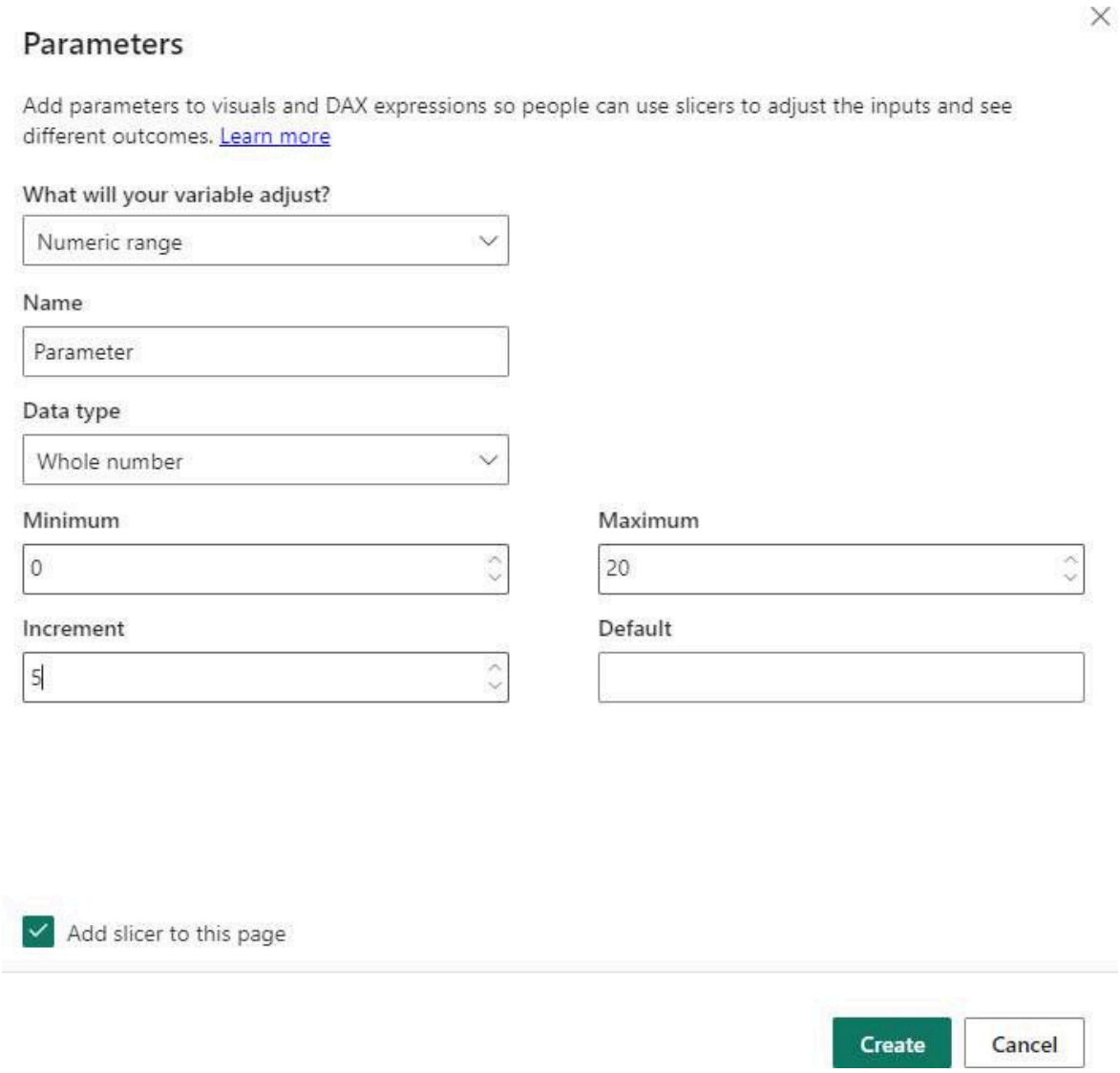
First, go to the tab “Modeling” and click on “New parameter”. Then choose “Numeric range”.

- If we have a strategy to set promotions for clients by offering discounts (such as 5%, 10%, 15%, and 20%), we will compare the number of “Sales” for this offer. Therefore, the “**Numeric range**” was chosen for that purpose.

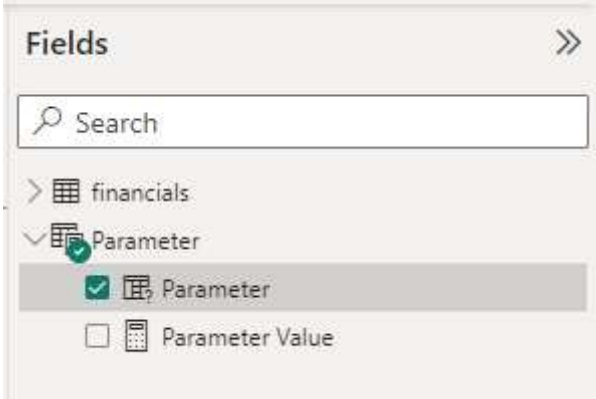


We can choose the value for each parameters below.

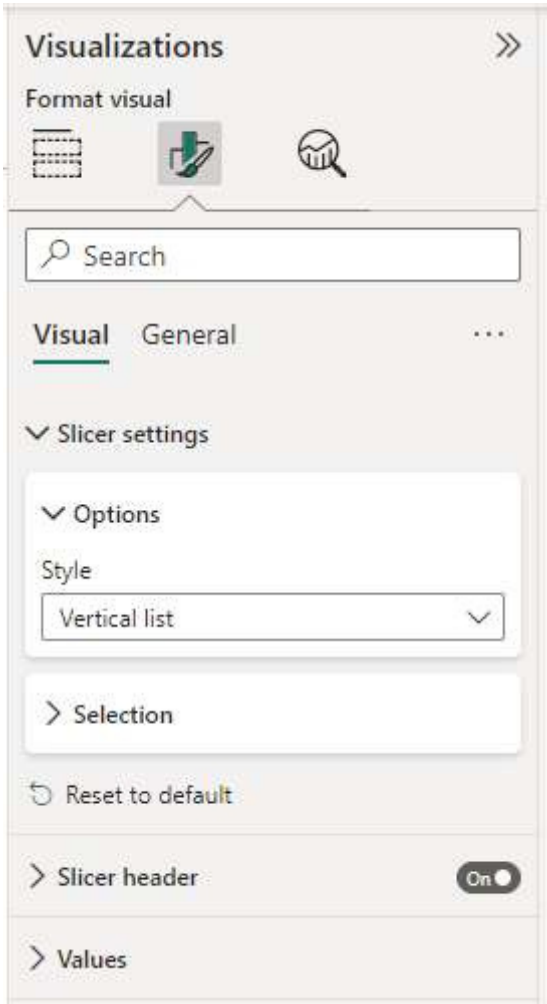
- What will your variable adjust? : For choosing an appropriate filter for your data.
- Name: To edit the table name as required.
- Data type: To choose the appropriate filter for your data.
- Minimum: To set a minimum value.
- Maximum: To set a maximum value.
- Increment: A series of regular consecutive additions.
- To display the slicer on canvas, check the "**Add a slicer to this page**" box.



After creating the parameter, we will **get the new table 'Parameter'** on "Fields".



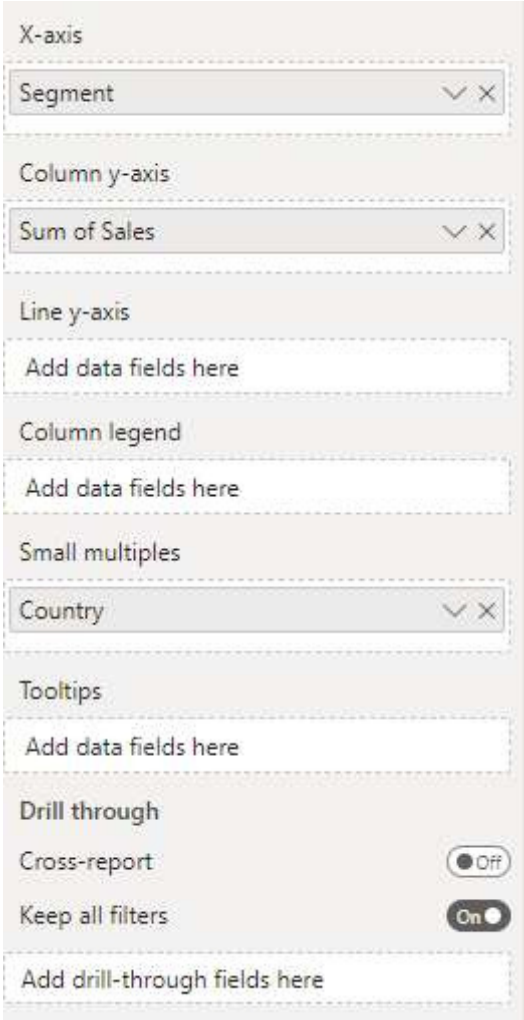
We can do this on the tab "Visual" of the menu "Format visual" on "Visualizations" and then change the field at "Slicer settings" on "Options" to **change the pattern slicer from "Single value" to "Vertical list."**



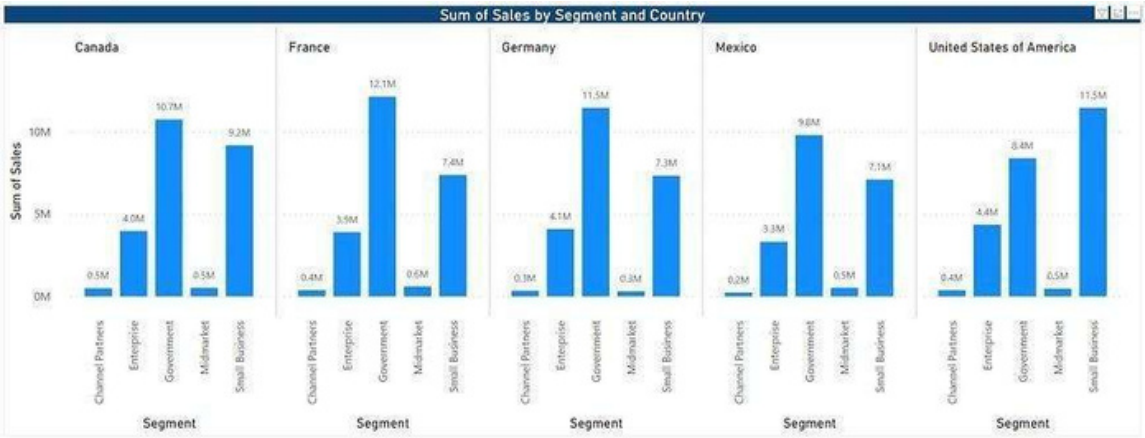
We will also need to create a measure to generate an interactive value and graph. Then go to the next step.

Step 2: Create a “measure”.

We create a visualization “Line and clustered column chart”. And fill value for each field as below.



We will get the chart as shown below.



Next, start to create a DAX measure

- **Parameter-SalesValue:** Create a measure on the “Parameter” table.

```
Parameter-SalesValue = SELECTEDVALUE('Parameter'[Parameter])
```

- **Sales Discount:** We create a new measure called “Sales Discount” that is calculated by multiplying the sum of Sales (on the “financials” table) by Parameter-SalesValue (on the “Parameter” table). Then, to convert the unit to a percentage, divide the result by 100.

```
Sales_Discount = ((sum(financials[ Sales])*Parameter[Parameter-SalesValue])/100)
```

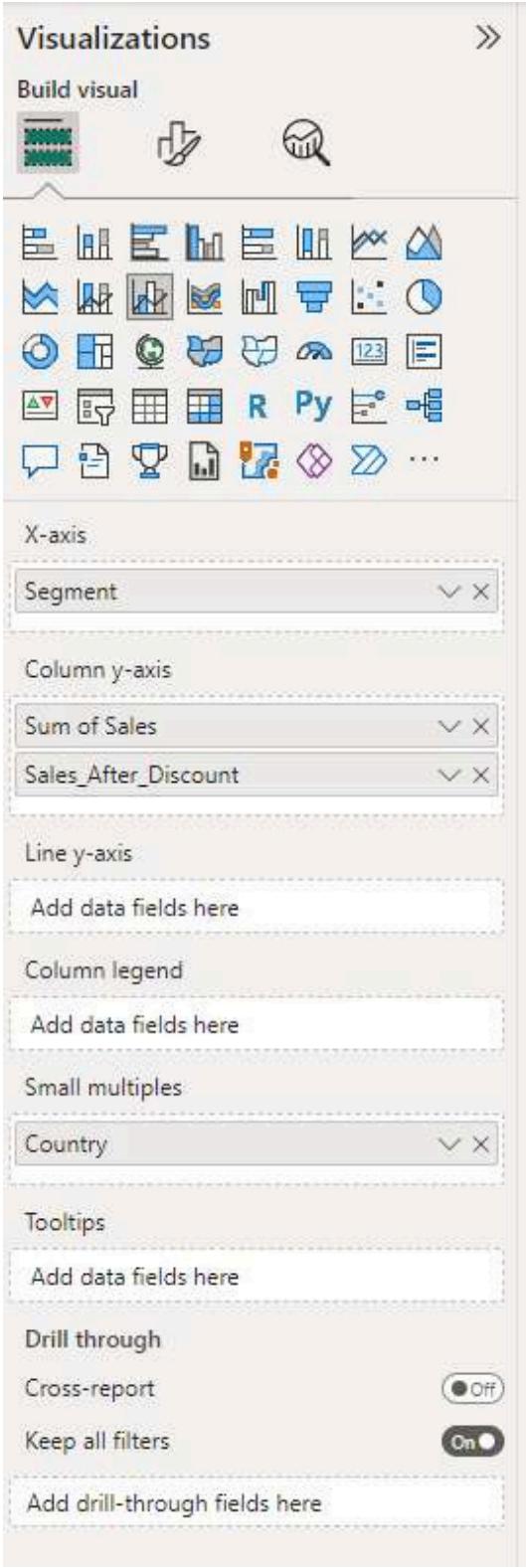
- **Sales After Discount:** This is a measure for calculating the formula that will show the result of the final price after getting the discount (refer to % discount on the parameter).

```
Sales_After_Discount = sum(financials[ Sales]) - Parameter[Sales_Discount]
```

Step 3: Visual for what-if analysis.

It’s time to visual our data !!

- Fill the column in each field as below. Don’t forget to choose “Line and clustered column char”.


Visualizations
Build visual
X-axis
Segment
Column y-axis
Sum of Sales
Sales_After_Discount
Line y-axis
Add data fields here
Column legend
Add data fields here
Small multiples
Country
Tooltips
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here

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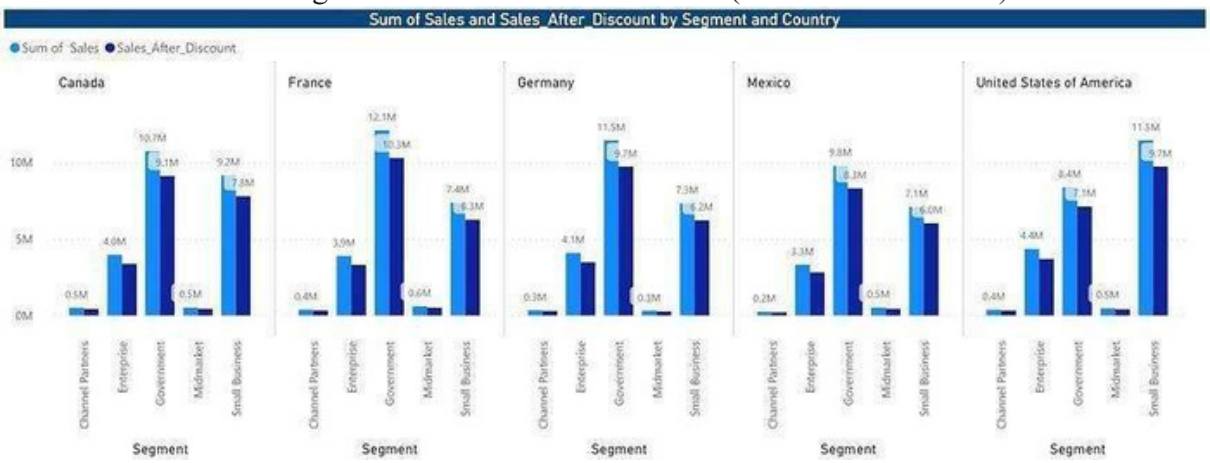
- Then we will get the bar chart is shown below (with a 15% discount).



- Finally, we get the report, which includes information about each country, segment, and discount percentage.

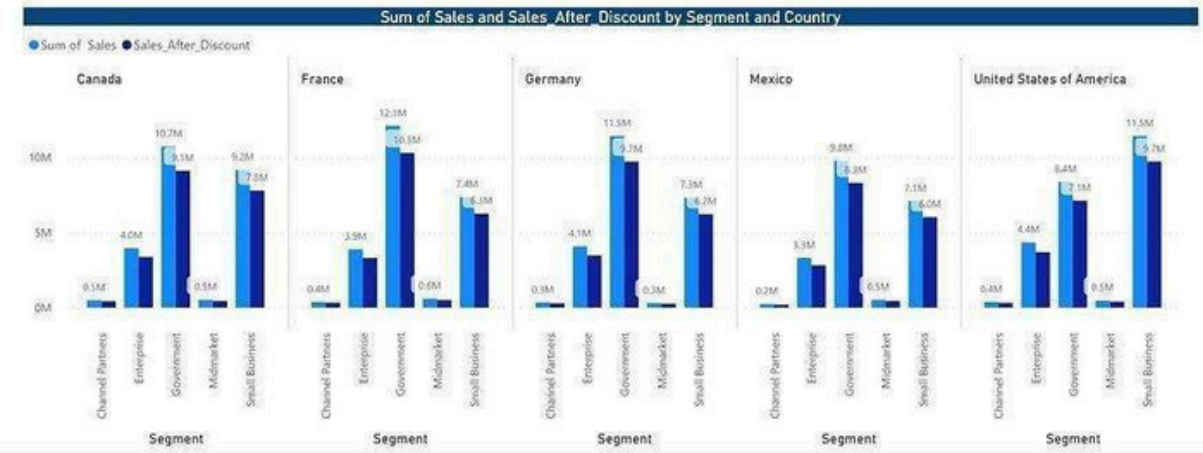
-

% discount

- ☐ 0
- ☐ 5
- ☐ 10
- ■ 15
- ☐ 20

Sales after get 15 % Discount

Country	Canada		France		Germany		Mexico		United States of America	
Segment	Sum of Sales	Sales_After_Discount	Sum of Sales	Sales_After_Discount	Sum of Sales	Sales_After_Discount	Sum of Sales	Sales_After_Discount	Sum of Sales	Sales_After_Discount
Channel Partners	491,164.14	417,489.52	372,090.36	316,276.81	336,425.88	285,962.00	234,379.08	199,222.22	366,534.18	311,554.05
Enterprise	3,967,491.25	3,372,367.56	3,890,890.63	3,307,257.03	4,086,826.25	3,473,802.31	3,315,881.25	2,818,499.06	4,350,605.00	3,698,014.25
Government	10,741,236.52	9,130,051.04	12,127,782.72	10,308,615.31	11,452,895.94	9,734,961.55	9,791,599.38	8,322,859.47	8,390,746.11	7,132,134.19
Midmarket	510,213.98	433,681.88	593,802.08	504,731.76	301,344.75	256,143.04	511,136.40	434,465.94	465,385.88	395,577.99
Small Business	9,177,549.00	7,800,916.65	7,369,606.50	6,264,165.53	7,327,848.00	6,228,670.80	7,096,356.00	6,031,902.60	11,456,559.00	9,738,075.15



RESULT:

Thus the What – If analysis for Data Visualization is applied to design and generate necessary on the data warehouse data in Excel successfully.